Ivan Osinnii, Ph.D.

Data Science Learner
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WORK EXPERIENCE

August 2024 – today
2 years

Postdoc – Novartis, Biomedical Research, Diseases of Aging and Regenerative Medicine

- Translational omics research, correlating genomic data with clinical outcomes
- Transforming bioinformatic pipelines with AI skills and agentic workflows
- Helped AI Product Lead with improving internal biomedical agent by feature suggestions
- Organized seminars on vibe coding for beginners in a BR community (mostly postdocs)

Jan 2023 – Feb 2024
1 year

Postdoc – Department of Biomedicine, University of Basel/University Hospital Basel

- Investigation of the role of Mucosal-Associated Innate-like T-cells in chronic liver disease
- Piloted single cell RNA-seq and IF image analyses, and contributed to grant application

May 2017 – Jun 2022
5 years

PhD student – Institute of Medical Microbiology, **University of Zurich**

- Pivoted to bioinformatics in the absence of department bioinformatician/mentor Publication of omics and in-vitro/in-vivo studies in ribosomal translation research
- Image analysis on confocal and wide-field microscopy of HEK293 (full pipeline)

Jan 2015 – Nov 2016
1 year

Research associate – Zabolotny Institute of Microbiology and Virology, Ukraine

- Performed survival analysis of in-vivo cohorts under experimental nanoparticle treatment

DATA SCIENCE/ML SKILLS

- Multivariate **linear/logistic regression** (non- or regularized), **LASSO & sPLS-DA**, **Bayesian** generative and compositional modeling (scCODA), **Tree-based Gene Regulatory Network inference** (SCENIC)
- Familiar with **loss function optimization**, **gradient descent**, data folding, tokenization, semantic geometry
- R language **functional programming**, **tidymodels**, **caret**, survival, lme4, **glmnet**
- **Python** – basic syntax, usage of **pandas, matplotlib** and **scikitlearn** packages
- Good command of data cleaning and preprocessing/transformation
- Working regularly on an upper end of medium size data on HPC cluster
- **Git and Github, UNIX command line**

AI/AUTOMATION SKILLS

- **LLMs** – tuning model performance with weights/context or temperature size, deep research, LLM Arena ranks
- **Agents** – **Claude Code CLI, VS Code Copilot**/Antigravity/Codex extensions, **autonomous cloud agents**
- **Fast prototyping** using parallelizing (parallel tmux sessions in composite repositories, or cloud agents)
- Creating agents in cloud platforms and programmatically using **LLM APIs** and specifications
- **.md Skills** with progressive disclosure for local and cloud agents (Manus, Biomni)
- GenAI Apps – Notion AI, **Gemini Notebook**, MS Copilot App/ Office plugins
- Beginner in local model inference

BIONFORMATICS / STATISTICS

- **Multiomics** – correlating several data modalities, like SomaScan proteomics, bulk and single cell RNA-seq, bulk RNA-seq and untargeted MS-based metabolomics
- **DGE, Seurat, AnnData objects** – single cell annotation, cell-cell interaction, batch effect removal
- Creating and evaluating a custom **Cox proportional hazards** model for biomarker discovery
- **Systems biology** analyses (PPI, GRNs, IPA, WGCNA)
- **Disease Target Identification** – basics (Open Targets, GTEx, Causaly)
- **Using Databases** – Uniprot, NCBI, MSigDB, Reactome etc.
- **Image analysis** – CellProfiler and QuPath

EDUCATION

May 2017 – Mar 2022	**PhD in Cell and Systems biology,** University of Zurich, Switzerland Thesis: The effects of mistranslation on proteostatic mechanisms in eukaryotic cells • Multiple transferable skill courses in a joint ETH-UZH graduate school
Sep 2013 – Dec 2014	**Master in Biotechnology** Thesis: Antiherpetic properties of ceria dioxide nanoparticles National University of Life and Environmental Sciences of Ukraine, Kyiv • Excellent grades: Diploma with distinction (GPA= 92/100)
Sep 2009 – Jun 2013	**Bachelor in Biotechnology** National University of Life and Environmental Sciences of Ukraine, Kyiv • Scholarship of the President of Ukraine for winning a scientific contest in plant genetics

- Business English and German**,** native Russian and Ukrainian languages

ACADEMIC ACHIEVEMENTS

- 2 manuscripts in preparation (one advanced)
- co-authored 3 manuscripts in biological journals with impact factor 5+

EXTRACURRICULAR ACTIVITY AND COURSES

- Member of a local Toastmasters club
- Writing popular scientific articles for educational platform scisteps.org and LinkedIn blog
- Course on Integration of Artificial Intelligence in Scientific Work, Blastim agency, 2024
- Coursera course “Introduction to the Biology of Cancer” with distinction, John Hopkins University, 2022

SOFT SKILLS

- Team play, scientific presentation and writing, establishing collaborations
- Troubleshooting, critical thinking, attention to detail
- Emotional intelligence, fast learner, punctuality

HOBBIES

- Playing guitar, hiking in mountains and doing yoga